ELEC2104/9704 Tutorial 9 Solutions (2025)

Introduction to MOSFETs and Circuit Analysis

The MOSFET: A Voltage-Controlled Current Source

The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is the fundamental building block of modern electronics. Unlike bipolar junction transistors (BJTs) which are current-controlled, MOSFETs are voltage-controlled devices - the gate voltage controls the drain current.

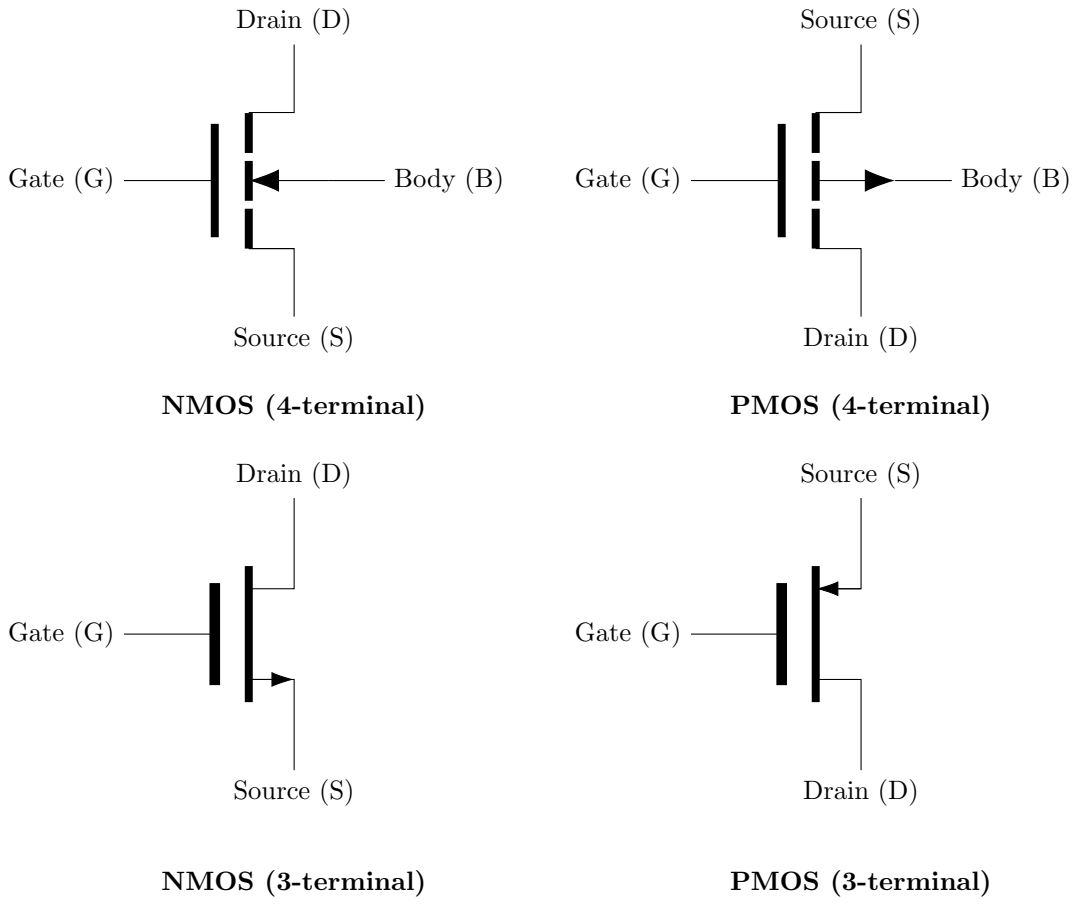


Figure 1: MOSFET symbols showing both 4-terminal (with body) and 3-terminal configurations

Critical: Arrow Direction Confusion Between 4-Terminal and 3-Terminal Symbols

The arrow direction differs between 4-terminal and 3-terminal MOSFET symbols!

Device	4-Terminal Arrow	3-Terminal Arrow
NMOS	INTO body terminal	OUT OF source terminal
PMOS	OUT OF body terminal	INTO source terminal

Understanding MOSFET Operation

The Physical Picture

When we apply a voltage to the gate (relative to the source), we create an electric field that attracts charge carriers to form a conductive channel between drain and source. This channel acts like a voltage-controlled resistor, allowing current to flow.

Key Parameters

- **Gate Current ($I_G = 0$)**: is zero (in this course), as the gate is insulated by oxide.
- **Threshold Voltage (V_{Th}, V_{TN} or V_{TP})**: Minimum gate voltage needed to form a channel
- **Transconductance Param. (K_n, K_p)**: $K' = \mu C_{ox}$ (μ carrier mobility, C_{ox} oxide capacitance)
- **Aspect Ratio (W/L)**: Width-to-length ratio of the channel, controlling current capability
- **Channel Length Modulation (λ)**: Accounts for slight dependence of I_D on V_{DS} in saturation

The Three Operating Regions

A MOSFET's behavior changes dramatically depending on the applied voltages. Understanding these regions is crucial for circuit analysis. The Triode region is split into linear and non-linear parts.

MOSFET Operating Regions

1. **Cutoff Region** (NMOS: $V_{GS} < V_{TN}$)
   - No channel formed, so acts like an open switch, $\mathbf{I_D = 0}$ (device is OFF)
   - PMOS have negative V_{GS} and V_{TP}, so the inequality is reversed: $V_{GS} > V_{TP}$
2. **Triode Region (Linear)** (NMOS: $V_{GS} > V_{TN}$ and $V_{DS} \ll 2(V_{GS} - V_{TN})$)
   - Channel formed and continuous
   - Current depends on both V_{GS} and V_{DS} linearly: $\mathbf{I_{D,tri} = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TN})V_{DS}}$
   - Acts like a voltage-controlled resistor: $\mathbf{R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TN})}}$
3. **Triode Region (Non-linear)** (NMOS: $V_{GS} > V_{TN}$ and $V_{DS} < V_{GS} - V_{TN}$)
   - The channel is still continuous but starts to pinch off near the drain
   - Nonlinear dependence on V_{GS}, V_{DS}: $\mathbf{I_{D,tri} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}\left[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2\right]}$
   - Acts like a voltage-controlled current source
   - PMOS: negative voltages (so reversed inequalities) $V_{GS} < V_{TP}$ and $V_{DS} > V_{GS} - V_{TP}$
4. **Saturation Region** (NMOS: $V_{GS} > V_{TN}$ and $V_{DS} \geq V_{GS} - V_{TN}$)
   - Channel pinched off near drain
   - Current mainly depends on V_{GS}: $\mathbf{I_{D,sat} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TN})^2(1 + \lambda V_{DS})}$
   - Acts like a voltage-controlled current source
   - PMOS: negative voltages (so reversed inequalities) $V_{GS} < V_{TP}$ and $V_{DS} \leq V_{GS} - V_{TP}$
   - **Remember this one!**

Don't confuse the BJT saturation region with the MOSFET saturation region! We use the MOSFET saturation region for amplification, while BJT saturation is for switching, not amplifying.

Analyzing MOSFET Circuits: The Load Line Method

When we place a MOSFET in a circuit with resistors and voltage sources, we need a systematic way to find the operating point. This is where load line analysis becomes powerful.

What is a Load Line?

The load line represents the constraint imposed by the external circuit. It's called a "load" line because it typically represents the load (resistance) connected to the MOSFET.
Consider this fundamental circuit:

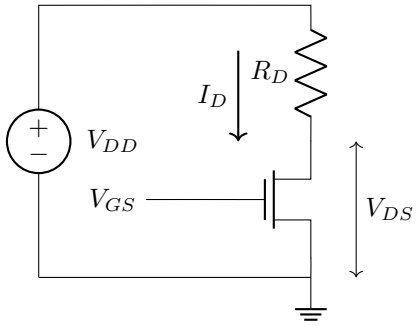


Figure 2: Basic MOSFET circuit for load line analysis

Applying Kirchhoff's Voltage Law (KVL) around the drain-source loop:

$$V_{DD} = I_D R_D + V_{DS} \tag{1}$$

Rearranging to get the load line equation:

$$I_D = \frac{V_{DD} - V_{DS}}{R_D} \tag{2}$$

On the I_D versus V_D plot, this is a straight line with:

- Y-intercept (when $V_{DS} = 0$): $I_D = \frac{V_{DD}}{R_D}$ (maximum possible current)
- X-intercept (when $I_D = 0$): $V_{DS} = V_{DD}$ (maximum possible voltage)
- Slope: $-\frac{1}{R_D}$ (determined by load resistance)

Graphical Solution: Where Device Meets Circuit

The beauty of load line analysis is that it provides a visual solution. The operating point (Q-point) is where the load line intersects the MOSFET characteristic curve.

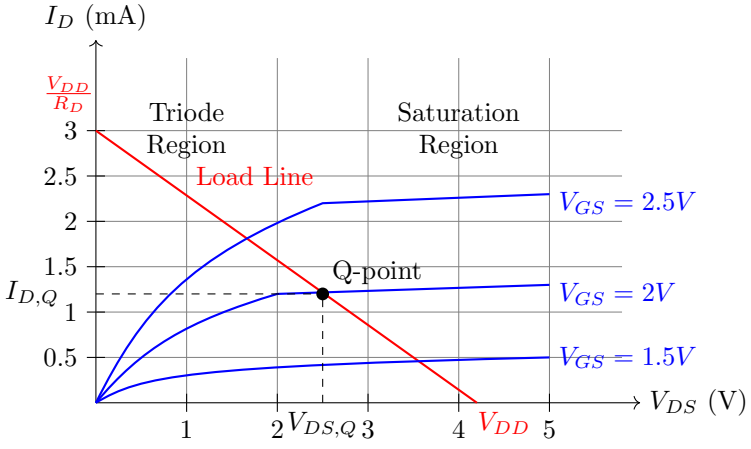


Figure 3: Load line intersecting with MOSFET characteristics to find Q-point

Solution Strategy

Systematic MOSFET Circuit Analysis

Step 1: Simplify the Circuit to get Load Line Equation

- Find Thévenin equivalents for bias networks if necessary (e.g. 4-Resistor Bias Network)
- Identify circuit constraints (KVL, KCL). E.g., KVL loops involving V_{DS} or V_{GS}.
- Use $I_G = 0$ to simplify analysis.

Step 2: Assume Operating Region to get MOSFET Equation

- For amplifiers: usually saturation (common in this course)
- Write the corresponding I_D equation for this region

Step 3: Solve Load Line and MOSFET Equations Simultaneously

- Combine MOSFET equation with circuit constraints (i.e. load line equation)
- Solve with the quadratic equation, keep the physically meaningful solution
  - Choose a solution that satisfies the region conditions in Step 4 (if there is one).

Step 4: Verify Assumption

- Check region conditions:
  - Saturation: $V_{GS} > V_{TN}$ and $V_{DS} > V_{GS} - V_{TN}$ (NMOS)
  - Triode: $V_{GS} > V_{TN}$ and $V_{DS} < V_{GS} - V_{TN}$ (NMOS)
- If wrong, return to Step 2 with new assumption

Step 5: Find Q-Point

- Calculate final values to get $Q = (I_D, V_{DS})$, returning to circuit analysis if necessary.
- This is your DC operating point

Advanced Topics

Channel Length Modulation

In saturation, the effective channel length decreases slightly with increasing V_{DS}, causing a small increase in current:

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_{TN})^2(1 + \lambda V_{DS}) \tag{3}$$

This makes the saturation curves slightly upward-sloping rather than perfectly flat.
Default: For most problems, we ignore λ (set $\lambda = 0$) unless specified otherwise.

Body Effect

MOSFETs actually have a 4th terminal called the "Body", which is typically just connected to the source and so is not shown (see Q2). However, when the source is not connected to the body (substrate), we must take the "Body Effect" into account, where the threshold voltage increases. For NMOS:

$$V_{TN} = V_{T0} + \gamma\left(\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F}\right) \tag{4}$$

For PMOS:

$$V_{TP} = V_{T0} - \gamma\left(\sqrt{V_{BS} + 2\phi_F} - \sqrt{2\phi_F}\right) \tag{5}$$

Note the negative sign before γ for PMOS! This is because increasing V_{BS} increases the *magnitude* of the threshold voltage (makes it more negative).
Where:

- V_{T0}: Zero-bias threshold voltage
- γ: Body effect parameter
- V_{SB} / V_{BS}: Source-to-body voltage / body-to-source voltage
- $2\phi_F$: Surface potential parameter ($2\Psi_F$ in the lectures)

This creates an implicit equation since $V_{SB} = I_D R_3$ in many circuits, requiring iterative solution.

Tip: You can check the *magnitude* of the threshold voltage for either NMOS and PMOS, with a unified formula:

$$|V_{Th}| = |V_{T0}| + \gamma \left(\sqrt{|V_{SB}| + 2\phi_F} - \sqrt{2\phi_F} \right) \tag{6}$$

Remember to make your answer for $|V_{Th}|$ negative if its PMOS (i.e. V_{TP}).

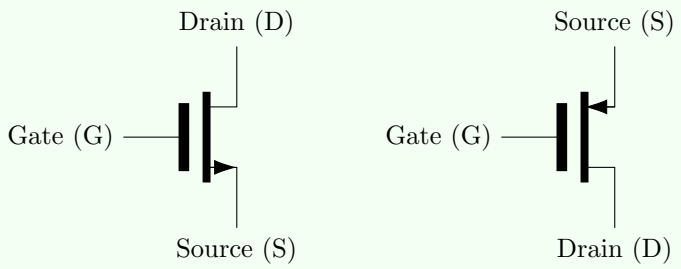
Threshold Voltage Notation

Important: Don't confuse MOSFET threshold voltages with BJT thermal voltage V_T!

- V_{TN} = NMOS threshold voltage (typically positive, e.g., +0.75 V)
- V_{TP} = PMOS threshold voltage (typically negative, e.g., -0.75 V)
- V_{Th} = MOSFET threshold voltage (generic symbol, either NMOS or PMOS)
- V_{Th0} or V_{T0} = Zero-bias threshold voltage (when $V_{SB} = 0$ or $V_{BS} = 0$)

Note: When there's no body effect, $V_{TN} = V_{T0}$ for NMOS and $V_{TP} = V_{T0}$ for PMOS.

Three-Terminal Symbol Convention



NMOS (3-terminal) **PMOS (3-terminal)**

Whenever you see a **3-terminal MOSFET symbol** (without an explicit body connection), it is implied that:

- The body is internally connected to the source
- Therefore, $V_{BS} = 0$ (for PMOS) or $V_{SB} = 0$ (for NMOS)
- There is no body effect (V_{TN} or V_{TP} remains constant)
- This is the most common configuration in basic circuits

Only when you see the **4-terminal symbol** with a separate body connection **and** $V_{SB} \neq 0$ (or $V_{BS} \neq 0$) **and** $\gamma \neq 0$, should you consider body effect!

Effect of W/L Ratio

The aspect ratio directly scales the current capability:

- Doubling W/L doubles the saturation current
- Allows designer to size transistors for required current
- Trade-off: larger devices consume more area and have higher capacitance

Design Insight

In the tutorial problems, notice how changing W/L from 6/1 to 12/1 (Question 4) shifts the Q-point along the load line. The load line stays fixed (determined by external resistors), but the MOSFET curves change, moving the intersection point.

Connecting Theory to Practice

As you work through the tutorial problems, you'll see these concepts in action:

1. **Question 1**: Self-biased configuration with load line analysis
2. **Question 2**: PMOS with body effect
3. **Question 3**: Transmission gate (parallel NMOS/PMOS)
4. **Questions 4-5**: Four-resistor bias with varying parameters

Each problem reinforces the fundamental principle: the Q-point is where the device physics (MOSFET equations) meets the circuit constraints (load line).

Summary: The Big Picture

MOSFETs are voltage-controlled devices that form the backbone of modern electronics. Understanding their operation requires:

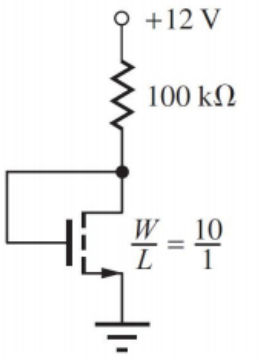
- Recognizing the three operating regions and their equations
- Using load line analysis to find the Q-point graphically
- Applying systematic solution strategies for different bias configurations
- Accounting for second-order effects when precision matters

The load line method bridges the gap between device physics and circuit behavior, providing both mathematical rigor and intuitive understanding. Master this technique, and you'll have a powerful tool for analyzing and designing MOSFET circuits.

1 Question 1

(a) Find the drain current for the transistor if $\lambda = 0$.

(b) Repeat if the W/L ratio is changed to 25/1.



Solution:

1.1 Part A

Note that $\lambda = 0$ means we are neglecting channel length modulation, simplifying our analysis by assuming the drain current is independent of V_{DS} in saturation.

Looking at the circuit, we can identify this as an NMOS transistor (the arrow on the source points outward). Another thing to note is that the gate is connected directly to the drain D, creating a self-biased configuration where $V_{GS} = V_{DS}$. This special configuration guarantees the transistor operates in saturation since:

$$V_{DS} = V_{GS} > V_{GS} - V_{TN} \text{ (always true as } V_{TN} \text{ is positive)}$$

For an NMOS in saturation with $\lambda = 0$, the drain current equation is:

$$I_D = \tfrac{1}{2}\mu_n C_{ox}\left(\tfrac{W}{L}\right)(V_{GS} - V_{TN})^2 \tag{1}$$

To find the load line equation, we apply KVL around the drain-source loop:

$$V_{DD} = I_D R_D + V_{DS}$$

where $V_{DD} = +12$ V and $R_D = 100\,\text{k}\Omega$. Solving for I_D gives Ohm's law:

$$I_D = \frac{V_{DD} - V_{DS}}{R_D}$$

Since the gate is connected to the drain in this circuit, we have $V_{DS} = V_{GS}$, therefore:

$$I_D = \frac{12\text{ V} - V_{DS}}{100\,\text{k}\Omega} = \frac{12\text{ V} - V_{GS}}{100\,\text{k}\Omega} \tag{2}$$

Equating equations (1) and (2), we can solve for V_{GS}:

$$\tfrac{1}{2}(\mu_n C_{ox})\left(\tfrac{W}{L}\right)(V_{GS} - V_{TN})^2 = \frac{12\text{ V} - V_{GS}}{100\,\text{k}\Omega} \tag{3}$$

This reduces to

$$12.5V_{GS}^2 - 17.8V_{GS} - 4.97 = 0 \tag{4}$$

The quadratic formula gives two solutions:

$$V_{GS} = 0.2\text{ V or } \ 1.2\text{ V} \tag{5}$$

We need to check which solution is physically valid. For the NMOS to be ON and in saturation:

- We need $V_{GS} > V_{TN} = +0.75$ V (transistor ON)
- The solution $V_{GS} = 0.2$ V $< V_{TN}$ would put the transistor in cutoff (OFF)
- Only the solution $V_{GS} = 1.2$ V $> V_{TN}$ is valid

Therefore, the operating point is $V_{GS} = 1.2$ V, giving:

$$I_D = \frac{12 \text{ V} - 1.2 \text{ V}}{100\,\text{k}\Omega} = \frac{10.8 \text{ V}}{100\,\text{k}\Omega} = 108\ \mu\text{A} \tag{6}$$

1.2 Part B

Following the same procedure, we assume saturation. Starting from the load line equation (2) and using the new aspect ratio $W/L = 25/1$:

$$\tfrac{1}{2}(\mu_n C_{ox})\left(\tfrac{25}{1}\right)(V_{GS} - V_{TN})^2 = \frac{12 \text{ V} - V_{GS}}{100\,\text{k}\Omega} \tag{7}$$

Simplifying this quadratic equation, we can get

$$62.5V_{GS}^2 - 91.75V_{GS} + 11.16 = 0 \tag{8}$$

Then the solutions are

$$V_{GS} = 0.46 \text{ V or } \ 1.046 \text{ V} \tag{9}$$

Again checking validity:

- $V_{GS} = 0.46$ V $< V_{TN} = 0.75$ V (cutoff - invalid)
- $V_{GS} = 1.046$ V $> V_{TN}$ (saturation - valid)

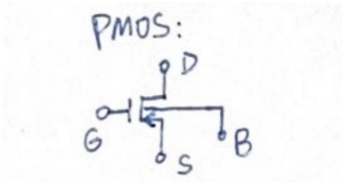
The operating point is $V_{GS} = 1.046$ V, giving:

$$I_D = \frac{12 \text{ V} - 1.046 \text{ V}}{100\,\text{k}\Omega} = \frac{10.954 \text{ V}}{100\,\text{k}\Omega} = 110\ \mu\text{A} \tag{10}$$

Note that increasing W/L from 6/1 to 25/1 increased the drain current slightly.

2 Question 2

A PMOS transistor is operating with $V_{BS} = 4$ V, $V_{GS} = -1.5$ V, and $V_{DS} = -4$ V. What is the region of operation and drain current in this device if $W/L = 25/1$?



PMOS Sign Conventions: For PMOS transistors, we use a different sign convention than NMOS:

- All voltages are typically negative relative to the source (which is usually at the highest potential),
- This is unlike BJT PNP transistors which use "flipped" subscripts (e.g., V_{EB} instead of V_{BE}), MOSFETs maintain consistent subscript ordering (e.g., V_{GS}, V_{DS}) for both NMOS and PMOS.
- Exception: For body effect, we *can* use V_{BS} (body-to-source) for PMOS (versus V_{SB} for NMOS).

Solution:

Since $V_{BS} = 4 \text{ V} \neq 0$, the body cannot be short-circuited to the source and thus we must include the body effect. The threshold voltage for the PMOS, V_{TP} is:

$$V_{TP} = V_{T0} - \gamma \left(\sqrt{V_{BS} + 2\phi_F} - \sqrt{2\phi_F} \right) \tag{1}$$

Using the PMOS parameters from the tutorial parameter section (note that PMOS typically has different parameters than NMOS):

- $V_{T0} = -0.75$ V (zero-bias threshold voltage for PMOS)
- $\gamma = 0.5\sqrt{\text{V}}$ (body effect parameter)
- $2\phi_F = 0.6$ V (surface potential parameter, $2\Psi_F$ in the lectures)

With PMOS we deal with a lot of negatives! Substituting these values:

$$V_{TP} = -0.75 \text{ V} - 0.5\sqrt{\text{V}} \left(\sqrt{4 \text{ V} + 0.6 \text{ V}} - \sqrt{0.6 \text{ V}} \right) = -1.44 \text{ V} \tag{2}$$

Now we check the region of operation to select the proper drain current formula. For PMOS:

- **ON condition:** $V_{GS} < V_{TP}$ (V_{GS} more negative than threshold)
- **Triode region:** $V_{DS} > V_{GS} - V_{TP}$ (V_{DS} less negative)
- **Saturation region:** $V_{DS} < V_{GS} - V_{TP}$ (V_{DS} more negative)

Checking our values:

$$V_{GS} = -1.5 \text{ V} < V_{TP} = -1.44 \text{ V} \quad \checkmark \text{ (ON)} \tag{3}$$

$$V_{DS} = -4 \text{ V} < V_{GS} - V_{TP} = -1.5 \text{ V} - (-1.44 \text{ V}) = -0.06 \text{ V} \quad \checkmark \text{ (Saturation)} \tag{4}$$

Therefore, the PMOS operates in the **saturation region**.

The saturation drain current formula for PMOS (without channel-length modulation effect $\lambda = 0$) is:

$$I_D = \tfrac{1}{2} K_p' \left(\tfrac{W}{L}\right) (V_{GS} - V_{TP})^2 \tag{5}$$

Substituting our values (note that $K_p' = \mu_p C_{ox}$ for PMOS):

$$I_D = \tfrac{1}{2}(40 \times 10^{-6} \text{ A/V}^2)(25)[(-1.5 \text{ V}) - (-1.44 \text{ V})]^2 \tag{6}$$

$$= \tfrac{1}{2}(40 \times 10^{-6} \text{ A/V}^2)(25)(-0.06 \text{ V})^2 \tag{7}$$

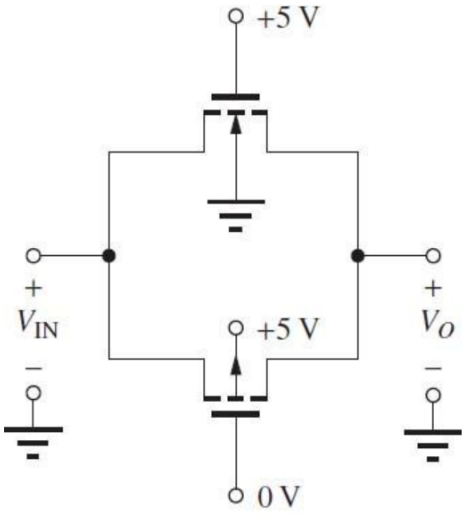
$$= 1.8 \ \mu\text{A} \tag{8}$$

3 Question 3

What is the on-resistance and voltage V_O for the parallel combination of the NMOS ($W/L = 10/1$) and PMOS ($W/L = 25/1$) transistors in a *transmission gate*?

(a) For $V_{IN} = 0$ V?

(b) For $V_{IN} = 5$ V?



Background: A transmission gate is a bidirectional switch that uses complementary NMOS and PMOS transistors in parallel. This configuration ensures good conduction for both high and low input voltages, overcoming the limitations of single transistor switches.

MOSFET On-Resistance: When a MOSFET operates in the triode (linear) region with small V_{DS}, it behaves like a resistor. The on-resistance R_{on} is the small-signal resistance for small drain-source voltages:

$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TN})}$$

MOSFET Symmetry

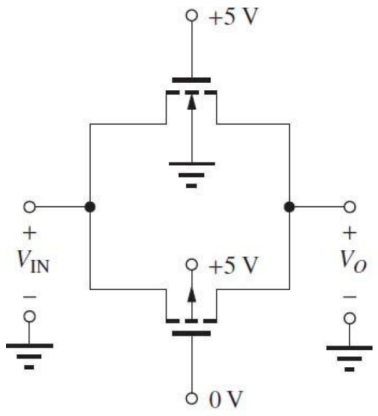
Unlike BJTs, MOSFETs are symmetric devices! This means:

- The drain and source terminals are physically identical
- You can swap drain and source, and the device works the same way
- The terminal at higher potential (for NMOS) becomes the "drain" (PMOS is opposite)
- The terminal at lower potential (for NMOS) becomes the "source" (PMOS is opposite)

For transmission gate analysis:

1. **Step 1:** Identify which terminal is at higher/lower potential
2. **Step 2:** Assign drain/source based on voltage levels
3. **Step 3:** Calculate V_{GS} using the correct source terminal
4. **Step 4:** Check if V_{GS} meets the ON condition

Example: If $V_{IN} = 0$ V and $V_{OUT} = 0$ V, both terminals are at same potential, so we can choose either as source for analysis.



Solution:

Part A

Given $V_{IN} = 0$ V, let's analyze the gate voltages (assuming $V_O \approx V_{IN} = 0$ V for the transmission gate):

- **NMOS (top):** $V_{GS} = V_G - V_S = 5\text{ V} - 0\text{ V} = 5\text{ V} > V_{TN} = 0.75\text{ V} \Rightarrow$ **ON**
- **PMOS (bottom):** $V_{GS} = V_G - V_S = 0\text{ V} - 0\text{ V} = 0\text{ V} > V_{TP} = -0.75\text{ V} \Rightarrow$ **OFF**

Therefore, only the NMOS is conducting, so the total resistance is:

$$R_{on} = R_{on,\text{NMOS}} = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TN})} \tag{1}$$

Substituting values:

$$R_{on} = \frac{1}{(100 \times 10^{-6}\text{ A/V}^2)(10/1)(5\text{ V} - 0.75\text{ V})} = \frac{1}{4.25 \times 10^{-3}\text{ A/V}} = 235\ \Omega \tag{2}$$

Since the transmission gate acts as a closed switch, $V_O = V_{IN} = 0$ V.

Part B

Given $V_{IN} = 5$ V, let's analyze the gate voltages (assuming $V_{OUT} \approx V_{IN} = 5$ V):

- **NMOS (top):** $V_{GS} = 5\text{ V} - 5\text{ V} = 0\text{ V} < V_{TN} = 0.75\text{ V} \Rightarrow$ **OFF**
- **PMOS (bottom):** $V_{GS} = 0\text{ V} - 5\text{ V} = -5\text{ V} < V_{TP} = -0.75\text{ V} \Rightarrow$ **ON**

Therefore, only the PMOS is conducting, so the total resistance is:

$$R_{on} = R_{on,\text{PMOS}} = \frac{1}{\mu_p C_{ox} \frac{W}{L}|V_{GS} - V_{TP}|} \tag{3}$$

Substituting values:

$$R_{on} = \frac{1}{(40 \times 10^{-6}\text{ A/V}^2)(25/1)|(-5\text{ V}) - (-0.75\text{ V})|} = \frac{1}{4.25 \times 10^{-3}\text{ A/V}} = 235\ \Omega \tag{4}$$

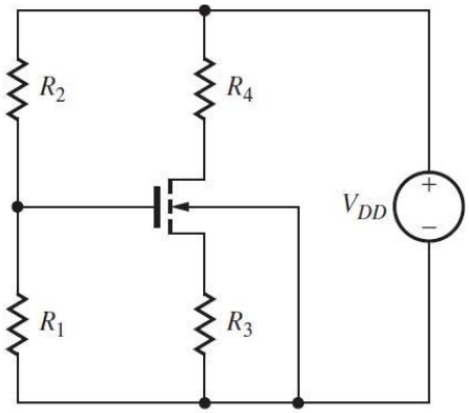
Since the transmission gate acts as a closed switch, $V_O = V_{IN} = 5$ V.

Notice that both cases yield the same on-resistance of 235 Ω. This is by design - the NMOS and PMOS are sized to provide equal resistances, ensuring consistent performance regardless of the input voltage level. This demonstrates the advantage of the transmission gate over single-transistor switches.

4 Question 4

(a) Find the Q-point for the transistor for $R_1 = 100$ kΩ, $R_2 = 220$ kΩ, $R_3 = 24$ kΩ, $R_4 = 12$ kΩ, and $V_{DD} = 10$ V. Assume that $V_{TO} = 1$ V, $\gamma = 0$, and $W/L = 6/1$.

(b) Repeat for $W/L = 12/1$.



Solution:

This problem uses the four-resistor bias configuration discussed in the introduction. This biasing method provides excellent temperature stability and is widely used in practical MOSFET circuits.

For a 4-resistor biased circuit, our first step is to convert the voltage divider (left part) to its Thévenin equivalent. The equivalent circuit diagram is drawn below.

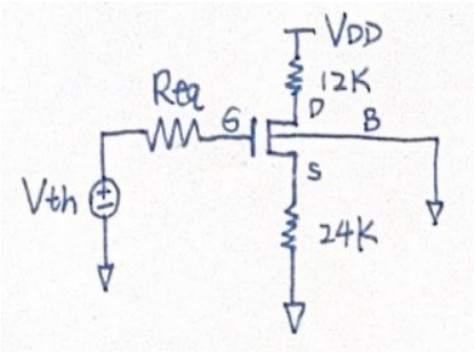


Figure 4: q4 equivalent circuit diagram

The following formulas are derived in the lectures. The Thevenin resistance is

$$R_{eq} = R_2 \parallel R_1 = 220\,k\Omega \parallel 100\,k\Omega = 68.75\,k\Omega \tag{1}$$

The Thevenin voltage is

$$V_{th} = V_{DD} \cdot \frac{R_1}{R_2 + R_1} = 10\,\text{V} \cdot \frac{100\,k\Omega}{220\,k\Omega + 100\,k\Omega} = 3.125\ \text{V} \tag{2}$$

Since $\gamma = 0$ (the body effect parameter), there is no body effect and the threshold voltage remains constant:

$$V_{TN} = V_{T0} = 1\ \text{V} \tag{3}$$

Note that the transistor is NMOS because the arrow on the body points inward.

4.1 Part A

For MOSFETs, the Q-point is conventionally expressed as (I_D, V_{DS}), representing the DC drain current and drain-source voltage.

We assume the NMOS operates in the saturation region. This is a reasonable assumption for amplifier circuits where we want the transistor to act as a voltage-controlled current source rather than a resistor.

Applying KVL around the gate-source loop (through the Thévenin equivalent):

$$V_{th} - I_G R_{eq} - V_{GS} - I_D R_3 = 0$$

Note that $I_G = 0\,\text{A}$, so the $I_G R_{eq}$ vanishes. Rearranging:

$$V_{th} = V_{GS} + I_D R_3 \tag{4}$$

According to our saturation assumption, the drain current equation is:

$$I_D = \tfrac{1}{2}\mu_n C_{ox}\left(\tfrac{W}{L}\right)(V_{GS} - V_{TN})^2 \tag{5}$$

Substituting equation (5) into (4):

$$V_{th} = V_{GS} + \left[\tfrac{1}{2}\mu_n C_{ox}\left(\tfrac{6}{1}\right)(V_{GS} - V_{TN})^2\right] R_3 \tag{6}$$

Simplifying gives the quadratic equation:

$$7.2V_{GS}^2 - 13.4V_{GS} + 4.08 = 0 \tag{7}$$

Solving yields one solution that satisfies $V_{GS} > V_{TN}$:

$$V_{GS} = 1.49\ \text{V} \tag{8}$$

Now the drain current is:

$$I_D = \tfrac{1}{2}(100 \times 10^{-6}\ \text{A/V}^2)(6)(1.49\ \text{V} - 1\ \text{V})^2 = 66.27\ \mu\text{A} \tag{9}$$

To find V_{DS}, we need to determine the drain and source voltages separately using KVL:

- **Drain voltage:** $V_D = V_{DD} - I_D R_4$ (voltage drop across R_4)
- **Source voltage:** $V_S = I_D R_3$ (voltage drop across R_3 from ground)

Therefore:

$$V_{DS} = V_D - V_S = \Big[V_{DD} - I_D R_4\Big] - \Big[I_D R_3\Big] \tag{10}$$

Substituting values:

$$V_{DS} = \big[10\ \text{V} - (66.27\ \mu\text{A})(12\,\text{k}\Omega)\big] - \big[(66.27\ \mu\text{A})(24\,\text{k}\Omega)\big] \tag{11}$$

$$= [10\ \text{V} - 0.795\ \text{V}] - [1.591\ \text{V}] = 7.62\ \text{V} \tag{12}$$

Next, we check the operation region.

$$V_{GS} > V_{TN}, \qquad V_{GS} - V_{TN} < V_{DS} \quad \Rightarrow \quad \text{saturation is correct } \checkmark \tag{13}$$

Finally, we write down the Q-point (quiescent or DC operating point):

$$Q = (I_D, V_{DS}) = (66.27\ \mu\text{A}, 7.62\ \text{V}) \tag{14}$$

4.2 Part B

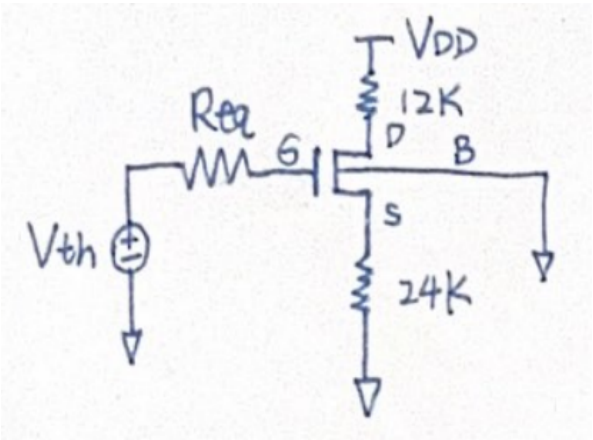


Figure 5: Q4 equivalent circuit diagram

Now we repeat the analysis with $W/L = 12/1$ instead of 6/1. Again, we assume saturation operation. The same KVL equation applies:

$$V_{th} = V_{GS} + I_D R_3 \tag{15}$$

But now the drain current equation has the new aspect ratio:

$$I_D = \tfrac{1}{2}\mu_n C_{ox}\left(\tfrac{12}{1}\right)(V_{GS} - V_{TN})^2 \tag{16}$$

Substituting the second equation into the first:

$$V_{th} = V_{GS} + \left[\tfrac{1}{2}\mu_n C_{ox}\left(\tfrac{12}{1}\right)(V_{GS} - V_{TN})^2\right] R_3 \tag{17}$$

Solving yields one solution that satisfies $V_{GS} > V_{TN}$:

$$V_{GS} = 1.35 \text{ V} \tag{18}$$

Then the drain current is:

$$I_D = \tfrac{1}{2}(100 \times 10^{-6} \text{ A/V}^2)(12)(1.35 \text{ V} - 1 \text{ V})^2 = 73.5 \ \mu\text{A} \tag{19}$$

The drain–source voltage calculation follows the same approach as Part A:

$$V_{DS} = V_D - V_S = \Big[V_{DD} - I_D R_4\Big] - \Big[I_D R_3\Big] \tag{20}$$

Substituting values:

$$V_{DS} = \big[10 \text{ V} - (73.5 \ \mu\text{A})(12\,\text{k}\Omega)\big] - \big[(73.5 \ \mu\text{A})(24\,\text{k}\Omega)\big] \tag{21}$$

$$= [10 \text{ V} - 0.882 \text{ V}] - [1.764 \text{ V}] = 7.35 \text{ V} \tag{22}$$

Next, we check the operation region.

$$V_{GS} > V_{TN}, \qquad V_{GS} - V_{TN} < V_{DS} \quad \Rightarrow \quad \text{saturation is correct } \checkmark \tag{23}$$

Finally, the Q-point for Part B is:

$$Q = (I_D, V_{DS}) = (73.5 \ \mu\text{A}, 7.35 \text{ V}) \tag{24}$$

Comparison: Notice that doubling the W/L ratio from 6/1 to 12/1 increased the drain current from 66.27 μA to 73.5 μA, while slightly reducing V_{DS} from 7.62 V to 7.35 V. This demonstrates how the aspect ratio affects the Q-point location along the load line.

5 Question 5

Find the Q-point for the transistor in the same four-resistor bias configuration as Question 4, with parameters: $R_1 = 100$ kΩ, $R_2 = 220$ kΩ, $R_3 = 24$ kΩ, $R_4 = 12$ kΩ, and $V_{DD} = 12$ V. Assume that $V_{T0} = 1$ V, $\gamma = 0.6$ V$^{1/2}$, and $W/L = 5/1$.

Solution:

Key Difference from Question 4: This problem is very similar to Question 4, but now includes the body effect ($\gamma \neq 0$ and $V_{SB} \neq 0\,\text{V}$) and also uses different supply voltage and aspect ratio.

The body effect makes the analysis more complex as the threshold voltage depends on the source voltage by $V_{TN} = V_{T0} + \gamma\left(\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F}\right)$.

Following the same approach as Question 4, we start by finding the Thévenin equivalent of the bias network:

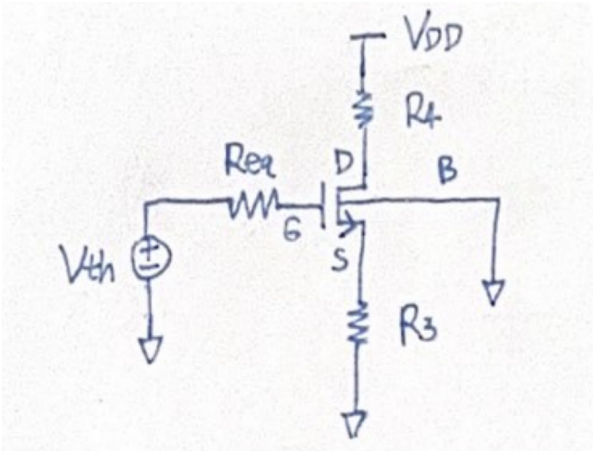


Figure 6: q5 equivalent circuit diagram

The Thévenin resistance (same calculation as Question 4):

$$R_{th} = R_{eq} = R_1 \parallel R_2 = \frac{100\,\text{k}\Omega \times 220\,\text{k}\Omega}{100\,\text{k}\Omega + 220\,\text{k}\Omega} = 68.75\,\text{k}\Omega \tag{1}$$

The Thévenin voltage (with the new supply voltage):

$$V_{th} = \frac{R_1}{R_1 + R_2} V_{DD} = \frac{100\,\text{k}\Omega}{320\,\text{k}\Omega} \times 12\text{ V} = 3.75\text{ V} \tag{2}$$

The threshold voltage V_{TN} including body effect is:

$$V_{TN} = V_{T0} + \gamma\left(\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F}\right) \tag{3}$$

Since $V_{SB} = V_S - V_B = I_D R_3 - 0 = I_D R_3$ (source above body), we have:

$$V_{TN} = 1\text{ V} + 0.6\sqrt{\text{V}}\left(\sqrt{I_D R_3 + 0.6\text{ V}} - \sqrt{0.6\text{ V}}\right) \tag{4}$$

From KVL around the gate-source loop (through the Thévenin equivalent):

$$V_{GS} = V_G - V_S = V_{th} - I_G R_{eq} - I_D R_3 = V_{th} - I_D R_3 \quad (\text{since } I_G = 0\,\text{A}) \tag{5}$$

Assuming the NMOS operates in saturation, the drain current equation is:

$$I_D = \tfrac{1}{2}\mu_n C_{ox}\left(\tfrac{W}{L}\right)(V_{GS} - V_{TN})^2 \tag{6}$$

This creates a system of implicit equations because V_{TN} depends on I_D through the body effect, and I_D depends on V_{TN} through the saturation equation. The problem becomes:

Find I_D and V_{TN} such that:

$$V_{TN} = 1\ \text{V} + 0.6\sqrt{\text{V}}\left(\sqrt{I_D \cdot 24\,\text{k}\Omega + 0.6\ \text{V}} - \sqrt{0.6\ \text{V}}\right) \tag{7}$$

$$I_D = \tfrac{1}{2}(100 \times 10^{-6}\ \text{A/V}^2)(5)[(3.75\ \text{V} - I_D \cdot 24\,\text{k}\Omega) - V_{TN}]^2 \tag{8}$$

This nonlinear system requires iterative solution (e.g., Newton's method). The solution yields:

$$I_D = 73.1\ \mu\text{A} \tag{9}$$

$$V_{TN} = 1.46\ \text{V} \tag{10}$$

The drain–source voltage is calculated as in previous problems:

$$V_{DS} = V_D - V_S = [V_{DD} - I_D R_4] - [I_D R_3] \tag{11}$$

$$= 12\ \text{V} - (73.1\ \mu\text{A})(12\,\text{k}\Omega) - (73.1\ \mu\text{A})(24\,\text{k}\Omega) \tag{12}$$

$$= 12\ \text{V} - 0.877\ \text{V} - 1.754\ \text{V} = 9.37\ \text{V} \tag{13}$$

Next, we verify our saturation assumption:

- $V_{GS} = V_{th} - I_D R_3 = 3.75\ \text{V} - (73.1\ \mu\text{A})(24\,\text{k}\Omega) = 1.996\ \text{V}$
- $V_{GS} = 1.996\ \text{V} > V_{TN} = 1.46\ \text{V}$ ✓(transistor ON)
- $V_{GS} - V_{TN} = 0.536\ \text{V} < V_{DS} = 9.37\ \text{V}$ ✓(saturation confirmed)

Finally, the Q-point is:

$$Q = (I_D, V_{DS}) = (73.1\ \mu\text{A}, 9.37\ \text{V}) \tag{14}$$

Body Effect Impact: Notice that the body effect increased the threshold voltage from 1 V to 1.46 V, reducing the effective gate drive and affecting the operating point compared to the idealized case in Question 4.